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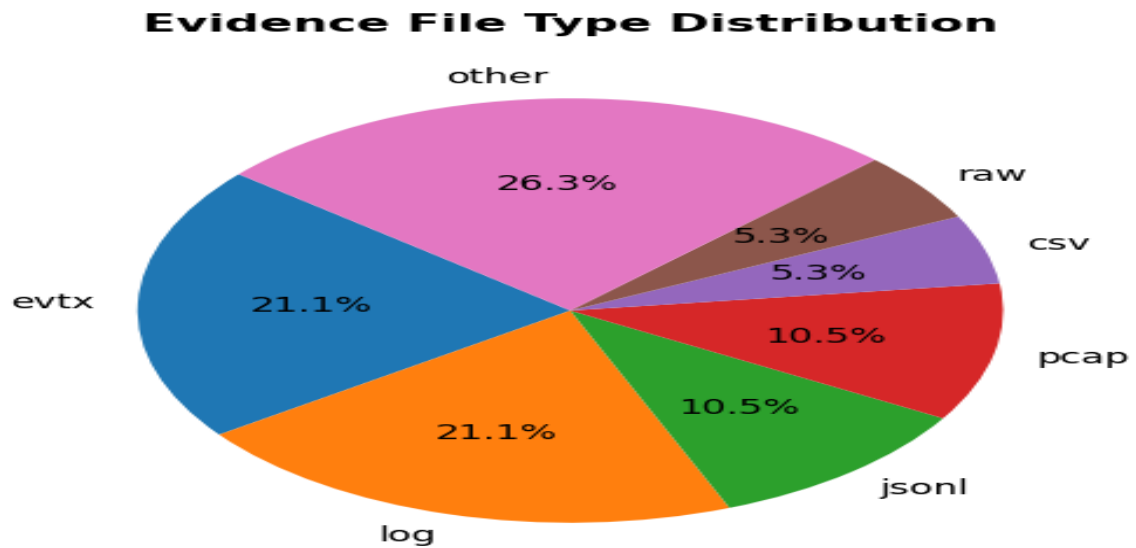
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## 1. Case Snapshot and Dataset Overview

Case CASE-2990-X42 was processed to summarize evidence intake, integrity, custody continuity, normalization quality, behavioral signals, and anomaly outcomes.

The dataset contains 19 artifacts with a combined size of 547.87 MB. The observed upload window spans from 2026-02-09 01:12:11 to 2026-02-09 04:05:27.

Most frequent formats in this dataset: evtx(4), log(4), jsonl(2), pcap(2), csv(1). These distributions help anticipate which parsers and correlation paths will contribute most to the investigation.

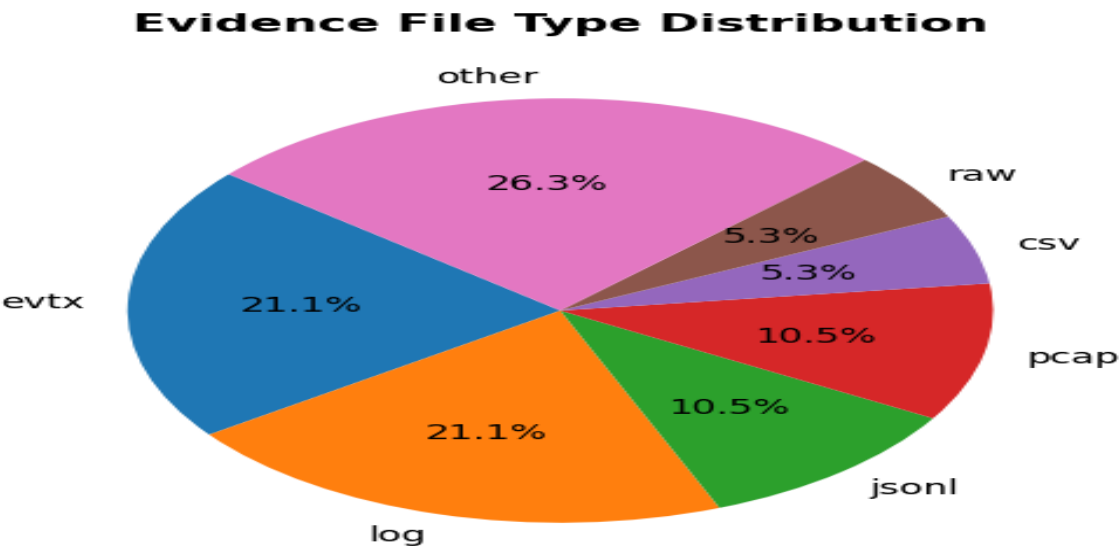


## 2. Evidence Intake Summary (Files, Formats, Sizes, Upload Times)

Evidence intake summarizes what was received, when it was received, and how it was categorized for downstream processing.

Artifacts are grouped by format and size to identify heavy sources (for example, memory dumps or PCAP segments) and to verify that the expected telemetry types are present.

The table below highlights the largest artifacts, which typically dominate processing time and often hold high-value forensic context.



### Largest Artifacts (Top 10)

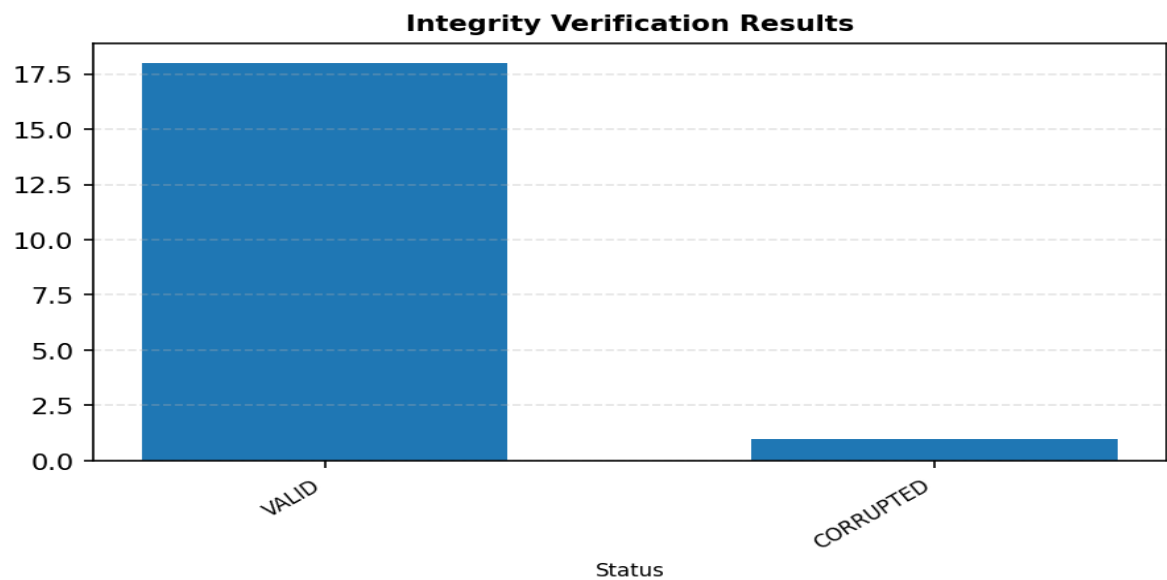
| Filename                          | Type  | Size      | Upload Time         |
|-----------------------------------|-------|-----------|---------------------|
| memory_dump_host07.raw            | raw   | 512.00 MB | 2026-02-09 03:10:44 |
| pcap_segment_20260209_002.pcap    | pcap  | 7.00 MB   | 2026-02-09 03:22:07 |
| pcap_segment_20260209_001.pcap    | pcap  | 6.00 MB   | 2026-02-09 03:18:22 |
| audit_export_20260209.zip         | zip   | 4.00 MB   | 2026-02-09 03:25:55 |
| edr_process_events_20260209.jsonl | jsonl | 3.00 MB   | 2026-02-09 02:24:10 |
| sysmon_20260209_0001.evtx         | evtx  | 2.34 MB   | 2026-02-09 01:19:22 |
| sysmon_20260209_0002.evtx         | evtx  | 2.20 MB   | 2026-02-09 01:21:40 |
| firewall_traffic_20260209.csv     | csv   | 2.00 MB   | 2026-02-09 02:18:06 |
| win_security_20260209_0001.evtx   | evtx  | 1.76 MB   | 2026-02-09 01:12:11 |
| win_security_20260209_0002.evtx   | evtx  | 1.64 MB   | 2026-02-09 01:13:58 |

### 3. Integrity Status (Per-File Hash Verification Results)

Per-file integrity verification ensures evidence remains bit-consistent from intake through processing.

Overall distribution: VALID=18, CORRUPTED/FAIL=1.

Any corrupted or failed artifacts should be isolated for re-acquisition or manual handling to avoid contaminating downstream analytics.



#### Integrity Status (First 20)

| Filename                          | SHA-256 (short) | Status    |
|-----------------------------------|-----------------|-----------|
| win_security_20260209_0001.evtx   | a3b9c1d5e7f2... | VALID     |
| win_security_20260209_0002.evtx   | b4c2d1e0f9a8... | VALID     |
| sysmon_20260209_0001.evtx         | c1d2e3f4a5b6... | VALID     |
| sysmon_20260209_0002.evtx         | d0e1f2a3b4c5... | VALID     |
| nginx_access_20260209_01.log      | e5f4d3c2b1a0... | VALID     |
| nginx_access_20260209_02.log      | f1a2b3c4d5e6... | VALID     |
| nginx_error_20260209.log          | 0a1b2c3d4e5f... | VALID     |
| firewall_traffic_20260209.csv     | 1b2c3d4e5f60... | VALID     |
| dns_queries_20260209.jsonl        | 2c3d4e5f6071... | VALID     |
| edr_process_events_20260209.jsonl | 3d4e5f607182... | VALID     |
| memory_dump_host07.raw            | 4e5f60718293... | VALID     |
| pcap_segment_20260209_001.pcap    | 5f60718293a4... | VALID     |
| pcap_segment_20260209_002.pcap    | 60718293a4b5... | VALID     |
| audit_export_20260209.zip         | 718293a4b5c6... | VALID     |
| host_inventory_20260209.xlsx      | 8293a4b5c6d7... | VALID     |
| suspicious_script_01.ps1          | 93a4b5c6d7e8... | CORRUPTED |
| webshell_suspect.aspx             | a4b5c6d7e8f9... | VALID     |
| linux_auth_20260209.log           | b5c6d7e8f90a... | VALID     |
| cloudtrail_20260209.json          | c6d7e8f90a1b... | VALID     |

4. Chain-of-Custody Status (Hash Chain Verification Results)

Chain-of-custody status summarizes whether custody continuity was preserved across the evidence lifecycle.

Hash chain status: VALID. Breaks observed: 0. First break: N/A.

If breaks are present, evidence after the first break should be treated as disputed until independently validated.

Hash Chain Summary

| Status | Breaks | First Break |
|--------|--------|-------------|
| VALID  | 0      | N/A         |

## 5. Parsing and Normalization Summary (Parsed vs Unparsed Records)

Parsing and normalization summarize how much raw telemetry could be converted into structured records ready for correlation.

Records: total=152400, parsed=149980, unparsed=2420.

Top parse error reasons: missing\_timestamp=880, bad\_json=610, unknown\_source\_type=540, encoding\_error=390. Addressing dominant error types typically improves coverage and reduces blind spots.

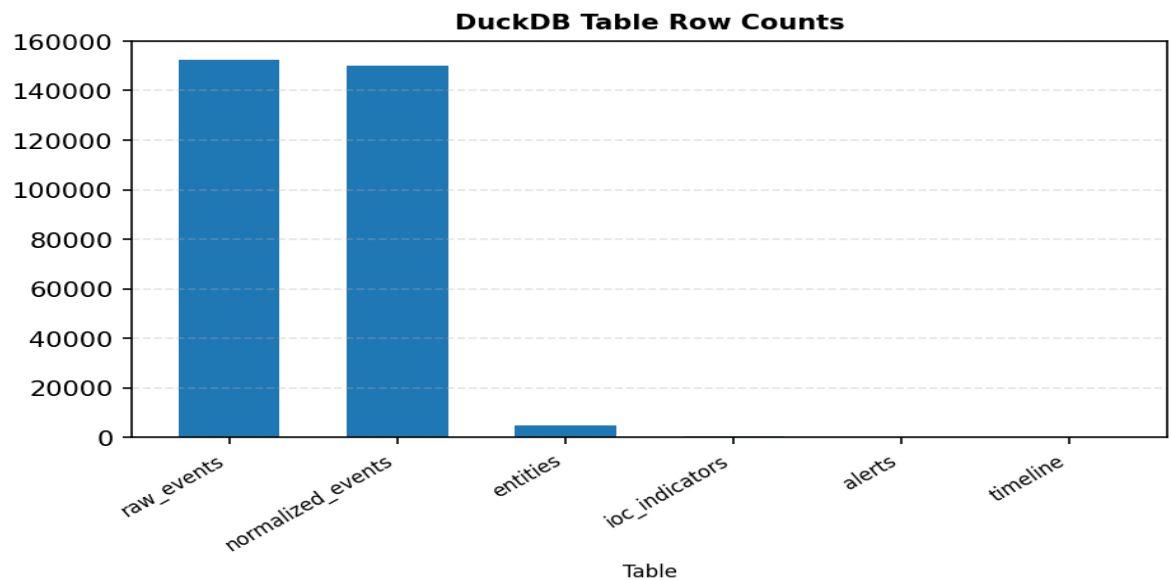


## 6. Hot Store Ingestion Summary (DuckDB Tables and Row Counts)

Hot store ingestion summarizes which normalized datasets were materialized for fast querying and analytics.

DuckDB table counts are a quick sanity check: they confirm that event streams, entities, indicators, and alerts were actually persisted after parsing.

If any expected table is missing or near-empty, it usually indicates an upstream parsing or mapping issue.



### DuckDB Materialization

| Table             | Rows   |
|-------------------|--------|
| raw_events        | 152400 |
| normalized_events | 149980 |
| entities          | 4870   |
| ioc_indicators    | 412    |
| alerts            | 186    |
| timeline          | 96     |

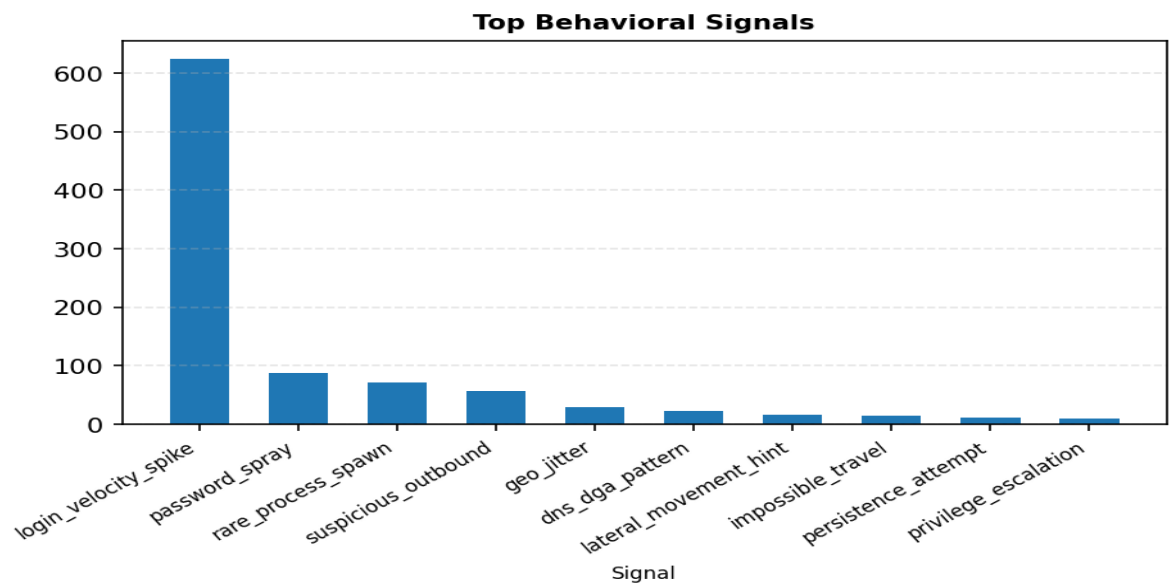


## 7. Behavioral Feature Summary (Signals Generated and Coverage)

Behavioral features describe what signals were computed from normalized records (for example: login velocity spikes, impossible travel, privilege escalations).

Signal counts give a quick view of coverage: higher counts often represent broad, low-confidence indicators; smaller counts can represent high-precision detections.

The chart below highlights the most frequent signals in this dataset.

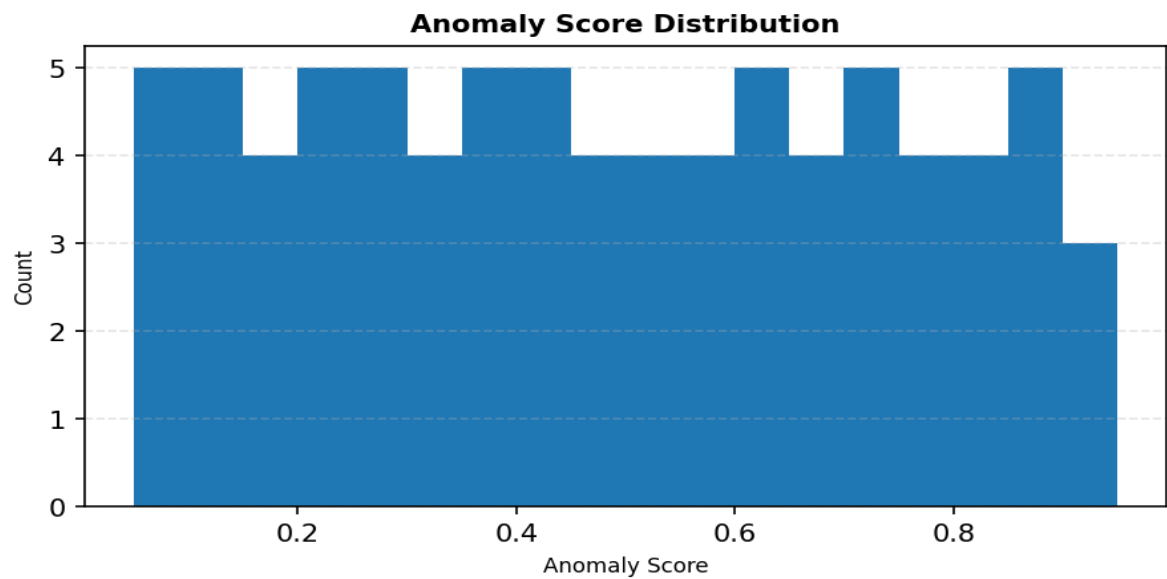


### 8. Anomaly Detection Summary (Total Scored, Anomalies Flagged, Score Range)

Anomaly detection summarizes how many records were scored and how many were flagged above the anomaly threshold.

Scored=149980, flagged=186, score range=0.03 to 0.98.

A tight score range usually indicates stable behavior; sharp peaks or long tails typically indicate bursty or staged activity.

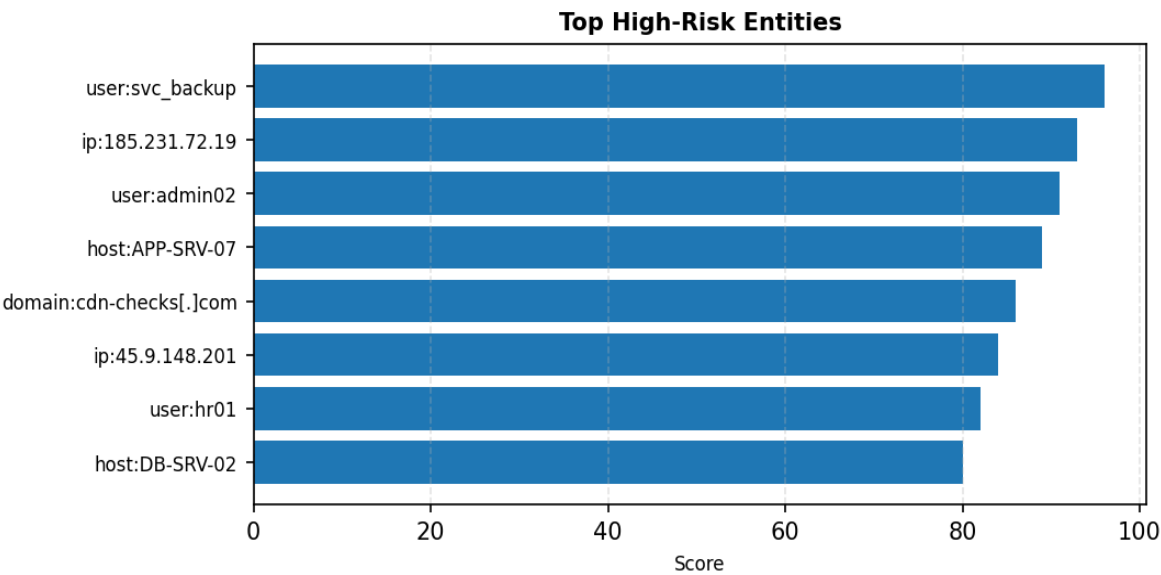


### 9. High-Risk Entities and Alerts (Users/IPs, Risk Levels, Key Triggers)

High-risk entities consolidate alerts by principal (user, host, IP, domain) and provide a ranked view of the most concerning actors.

A small number of entities frequently account for most high-confidence risk, making them good candidates for triage and containment checks.

The table below lists the highest-risk items and the triggers that caused scoring.



#### Top Alerts (First 15)

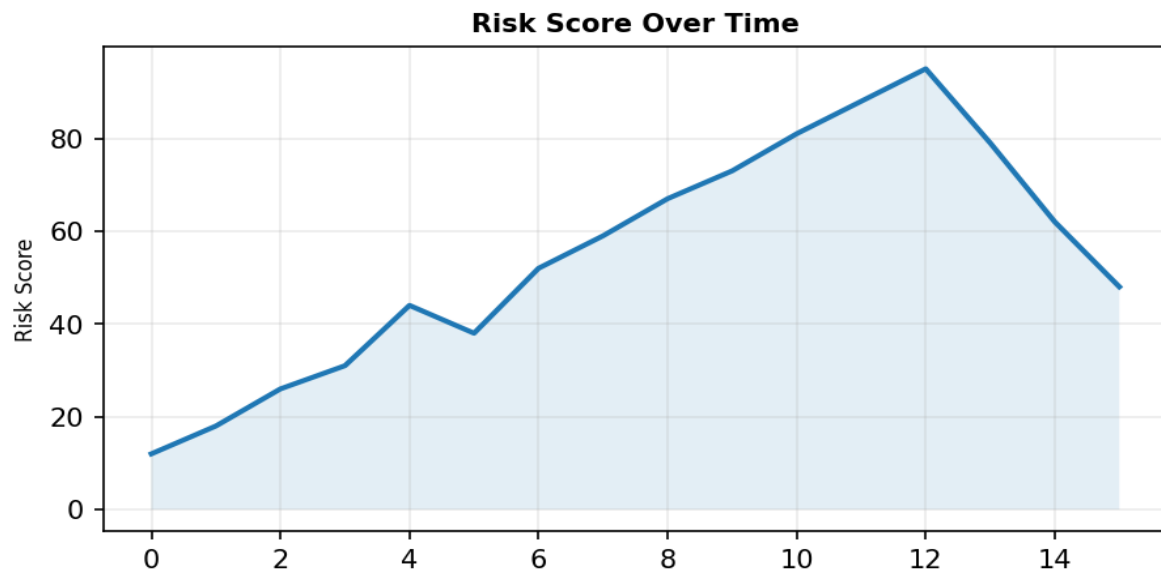
| Entity                  | Risk Score | Risk Level | Key Triggers                                     |
|-------------------------|------------|------------|--------------------------------------------------|
| user:svc_backup         | 96         | CRITICAL   | password_spray, privilege_escalation, log_tamper |
| ip:185.231.72.19        | 93         | HIGH       | suspicious_outbound, dns_dga_pattern             |
| user:admin02            | 91         | HIGH       | login_velocity_spike, impossible_travel          |
| host:APP-SRV-07         | 89         | HIGH       | rare_process_spawn, persistence_attempt          |
| domain:cdn-checks[.]com | 86         | HIGH       | dns_dga_pattern, suspicious_outbound             |
| user:hr01               | 82         | MEDIUM     | geo_jitter, login_velocity_spike                 |
| ip:45.9.148.201         | 84         | MEDIUM     | suspicious_outbound                              |
| host:DB-SRV-02          | 80         | MEDIUM     | large_data_transfer, suspicious_outbound         |

## 10. Investigation Timeline Highlights and Recommended Actions

Timeline highlights summarize how risk evolved over the observed window and identify the most important time slices for deeper artifact correlation.

Peak windows are the best starting point for reconstruction: pivot from the timestamp into supporting logs, process events, DNS, and network captures.

Recommended actions are derived from the dominant triggers observed in signals and entity alerts.



### Peak Windows (Top 5)

- 2026-02-09 03:27:44 — risk 95 — Peak anomaly wave; high-confidence suspicious activity window.
- 2026-02-09 03:12:55 — risk 88 — Multiple synchronized high-risk triggers correlate to same window.
- 2026-02-09 03:01:42 — risk 81 — Log tamper signals detected (sudden gaps / truncation behavior).
- 2026-02-09 03:42:10 — risk 79 — Outbound connections reduced after containment-like behavior.
- 2026-02-09 02:45:10 — risk 73 — Large data transfer spikes from DB-SRV-02 to external endpoint.

### Recommended Actions

- Enforce password reset for affected accounts; enable MFA for all privileged users; review lockout policy.
- Audit recent privilege changes and group membership updates; rotate admin credentials; validate sudo/admin use.
- Block suspicious domains at resolver and perimeter; check endpoints for beaconing; review DNS logs for related subdomains.
- Inspect outbound traffic for data exfil paths; quarantine endpoints showing repeated high-volume transfers; validate destinations.
- Review scheduled tasks/services/startup entries; inspect script execution history; collect EDR triage for impacted hosts.
- Preserve original logs; verify collector integrity; compare against upstream sources; expand timeline window for gaps.
- Document all actions with timestamps and operator identity to preserve auditability.